MIL-P-60169B
12 June 1970
SUPERSEDING
MIL-P-60169A
20 May 1969

MILITARY SPECIFICATION

PAPER, SEALING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers paper for foiling and sealing primer mixtures in primers.

1.2 Classification. Paper covered by this specification shall be of the following types as specified:

Type I - Paper coated, on one side only, with lacquer containing nitrocellulose.

Type II - Uncoated foiling paper (finish 50).

Type III - Foiling paper coated, on one side only, with shellac or suitable shellac substitute.

Type IV - Uncoated foiling paper (machine finish).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

UU-P-31 - Paper, General Specification and Methods of Testing.

FSC 1305

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:14Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

(Copies of specifications, standards, drawings and publications required by the supplier with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 The stock for type I and type II paper shall not contain ground wood in excess of ten percent; the remainder shall be chemical wood pulp.

3.1.2 The stock from which type III and type IV paper shall be made is optional.

3.2 Acidity. The pH value of the type I and type II paper shall be not less than 4.0.

3.3 Lacquer coating. The lacquer coating for type I paper shall contain nitrocellulose and shall conform to the requirements of table I. Manufacturers should certify that the lacquer used for coating type I paper meets the requirements of table I.

Table I

Total solids, percent by weight of lacquer	25 min.
Nitrocellulose content, percent of lacquer	14 ± 2
Weight per gallon	6.5 lbs., min.
Viscosity	10 poises
Nitrogen content of nitrocellulose, percent	2.0 min.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:14Z
Check the source to verify that this is the current version before use.

3.4 Color. White to buff, red to dark red or green to dark green, as specified in the contract or purchase order.

3.4.1 Color fastness. The color or red to dark red and green to dark green materials shall be fast to alcohol and acetone.

3.5 Finishes.

3.5.1 The type I and type II paper shall have a finish of approximately 50.

3.5.2 The type III and type IV paper shall have a machine finish but shall not have a glazed or heavily calendered surface.

3.6 Form.

3.6.1 The type I and type II paper shall be furnished in rolls as specified in the contract or order.

3.6.2 The types III and IV paper shall be furnished in roll or strip form as specified in the contract or order.

3.7 Ash. The ash content of the material shall not exceed five percent.

3.8 Combustibility. Type I paper shall burn completely in less than 50 seconds, when tested as described in 4.4.6.1. Types II, III and IV paper shall burn completely to ash, when tested as indicated in 4.4.6.2.

3.9 Basis weight. The basis weight of the paper shall be as follows:

Type I - 40 pounds plus or minus 2 pounds of base weight of paper coated with 12 pounds plus or minus 2 pounds of lacquer coating, as formulated in 3.3, per ream of 500 sheets 24 by 36 inches.

Type II - 40 pounds plus or minus 2 pounds of base weight per ream of 500 sheets 24 by 36 inches.

Type III - 54 plus or minus 2 pounds per ream of 500 sheets 24 by 36 inches.

Type IV - 42 plus or minus 2 pounds per ream of 500 sheets 24 by 36 inches.

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:14Z
Check the source to verify that this is the current version before use.

3.10 Dimensions. Unless otherwise specified, the paper shall be of the following dimensions.

3.10.1 Thickness.

Types I and III - 0.0040 inch plus or minus 0.0005 inch.
Types II and IV - 0.0035 inch plus or minus 0.0005 inch.

3.10.2 Width.

Types I and II - 2 3/16 inches minimum, 2 7/8 inches minimum or 4 inches minimum as specified
Types III and IV - 2 7/8 inches minimum or 4 inches minimum as specified.

3.10.3 Length.

3.10.3.1 Types I and II. Rolls having sufficient length to form a roll 8 to 9 inches in diameter having a core with inside diameter of 3 1/4 inches unless otherwise specified in the contract or order.

3.10.3.2 Types III and IV. Strips 51 inches long, corners bevelled, one end only, at 45 degree angle, 3/4 inch from each edge. Rolls having 8 to 15 pounds, as specified in the contract or order.

3.11 Comparison samples. With regard to finish, formation, cleanliness, pinholes, and stiffness, types III and IV paper shall be equal to comparison samples furnished with the invitation for bids.

3.12 Paper strength, types I and II.

3.12.1 Bursting strength. The bursting strength of types I and II paper shall be not less than 20 pounds.

3.12.2 Tearing resistance. The tearing resistance in each direction shall be not less than 40 grams.

3.12.3 Tensile strength. The tensile strength shall be as follows:

Machine direction, dry	10 lbs. per inch width, min.
Across machine direction, dry	6 lbs. per inch width, min.

3.13 Workmanship.

3.13.1 The stock for types I and II paper shall be uniformly beaten and refined so that a short fiber is produced and the machine direction of the fibers shall be as negative as possible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.

3.13.2 No loading materials, such as clay, shall be introduced in the manufacture of the paper.

3.13.3 The paper shall be uniform, free from pinholes, microholes, tears, creases, cuts, or other imperfections which may impair its usefulness.

3.13.4 Type I or type III paper shall not stick together as to cause tearings when unrolled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.3.2 Examination. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects contained herein using applicable sampling plans and acceptance criteria of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:14Z
Check the source to verify that this is the currentversion before use.

4.3.2.1 Rolls, prior to packing.

Defect	Method of inspection
CRITICAL	
None defined	
MAJOR AQL 0.25 percent	
101. Width	Measure
102. Thickness	Measure
103. Size of rolls	Measure
MINOR AQL 1.00 percent	
201. Evidence of poor workmanship	Visual

4.3.2.2 Rolls, after packing.

Defect	Method of inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR AQL 1.00 percent	
201. Improperly protected in shipping container	Visual
202. Marking incorrect, misleading or unidentifiable	

4.3.3 Sampling. Five rolls of paper shall be selected at random and shall be used to make up the test samples representatives of the lot. When the lot consists of less than five rolls, all the rolls in the lot shall be used to obtain the test samples. A two yard length of paper the full width of the roll shall be taken from each selected roll and used as a composite sample for the following tests listed in table II. Failure of a sample to comply with the applicable requirements in any test shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:14Z
Check the source to verify that this is the currentversion before use.

Table II

Test	Sample size and test procedures
Fiber	Method 300 - UU-P-31
Acidity	Method 200 - UU-P-31
Nitrocellulose coating for paper	Paragraph 4.4.2
Color fastness	Paragraph 4.4.5
Finish	Method 130 - UU-P-31
Ash	Method 202 - UU-P-31
Combustibility	Paragraph 4.4.6
Basis weight	Method 111 - UU-P-31
Thickness	Method 173 - UU-P-31
Bursting strength	Method 112 - UU-P-31
Tearing resistance	Method 170 - UU-P-31
Tensile strength	Method 171 - UU-P-31

4.4 Test methods and procedures.

4.4.1 Fiber identification. The fiber shall be determined in accordance with method 300 of UU-P-31.

4.4.2 Nitrocellulose content of lacquer. Using a weighing buret, slowly add approximately 2 grams of the sample to a 250 ml. beaker containing a stirring rod and 75 ml. of 65 percent acetic acid (by weight). During this addition, be sure that the sample drips directly into the acid solution and does not come in contact with the beaker wall or stirring rod; stir the mixture vigorously with the stirring rod to aid the precipitation of the nitrocellulose and prevent the formation of lumps. Achievement of proper extraction requires that the precipitate be loose and fibrous. Calculate the exact weight of the sample by difference. Cover the beaker with a tight-fitting (unribbed) watch glass and heat for 2 hours on a steam bath. Decant the hot solution through a tared sintered glass crucible of medium porosity with the aid of suction. Add 30 ml. of 65 percent acetic acid to the residue in the beaker, heat on a steam bath for 15 minutes, and decant through the crucible. Repeat the heating and decantation with 30 ml. more of 65 percent acetic acid and then with 30 ml. of water. Quantitatively transfer the nitrocellulose to the crucible with aid of a policeman and a stream of hot water and then wash the crucible several times with hot water. Dry the crucible at 100 ± 5°C for 2 hours, cool in a desiccator, and weigh the percentage of nitrocellulose as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.

$$\text{Percentage nitrocellulose} = \frac{(A-B)\ 100}{W}$$

where: A = weight of crucible and contents.
B = weight of crucible
W = weight of sample

4.4.2.1 Percent solids. Determine in accordance with method 4041 of Fed. Test Method Std. No. 141.

4.4.2.2 Weight per gallon. Determine in accordance with method 4184 of Fed. Test Method Std. No. 141.

4.4.2.3 Viscosity. Determine in accordance with method 4271 of Fed. Test Method Std. No. 141.

4.4.2.4 Nitrogen content of nitrocellulose. Determine by the nitrometer method or titanous chloride titration method in accordance with methods 209-3.2 and 209.7 of MIL-STD-286B.

4.4.3 Finish. The finish shall be determined in accordance with method 130 of UU-P-31.

4.4.4 Ash. Determine in accordance with method 202 of UU-P-31.

4.4.5 Color fastness. Immerse one portion of approximately 10 square inches of the sample in 25 ml. of alcohol and another in 25 ml. of acetone, both solvents being at room temperature. Note if the color fastness of the sample is satisfactory as indicated by the solvent remaining with only a slight discoloration after 15 minutes contact with the sample.

4.4.6 Combustibility.

4.4.6.1 Type I. Determine the burning rate by use of the apparatus shown in figure 1. Unscrew the bolts and insert a sample, 9 inch by 3 inch, in the fixture, hold a flame of a match at the top outside corner for 2 seconds, and note the time in seconds required for the flame to traverse the length of the paper, counting the time from the removal of the match. Perform the determination five times and calculate the average.

4.4.6.2 Types II, III and IV. Ignite a portion of the sample by means of the flame of a match and note if the combustibility of the sample is satisfactory as indicated by its complete combustion to ash.

4.4.7 Basis weight. The contractor shall furnish a certified report of the basis weight determined in accordance with method 110 of UU-P-31.

4.4.7.1 Type I only. Determination of the basis weight for type I shall be made before and after the extraction of the nitrocellulose lacquer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.

4.4.8 Atmospheric conditions. All the following physical tests on the paper shall be made on samples that have been conditioned in an atmosphere determined in accordance with method 102 of UU-P-31.

4.4.8.1 Bursting strength. Determine in accordance with method 112 of UU-P-31.

4.4.8.2 Tearing resistance. Determine in accordance with method 170 of UU-P-31.

4.4.8.3 Tensile strength. The tensile strength shall be determined in accordance with method 171 of UU-P-31.

4.4.9 Dimensions. Determine the required length and width by an convenient means accurate to at least 0.1 inch.

4.4.10 Thickness. Determine in accordance with method 111 of UU-P-31.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. Unless otherwise specified, the paper shall be packed in accordance with manufacturer's commercial practice to assure acceptance by common carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of foiling paper required. (See 1.2).
(c) Color of foiling paper required. (See 3.4).
(d) Form in which foiling paper is to be furnished. (See 3.6).
(e) Width in which paper is to be furnished. (See 3.10.2).
(f) State the required outside diameter, inside core diameter, and weight of the roll of paper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.

6.2 Supersession. This specification incorporates the requirements as indicated below.

MIL-P-60169B	MIL-P-60169A	MIL-P-60169(MU)	JAN-P-223
Type I	Type I	Type I	———
Type II	Type II	Type II	———
Type III	Type III	———	Type I
Type IV	Type IV	———	Type II

Custodian:
Army - MU
AF - 70

Review activities:
Army - MU
Air Force - 70

User activities:
Army - MI
Navy - OS

Preparing activity:
Army - MU

Project No. 1305-0626

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.

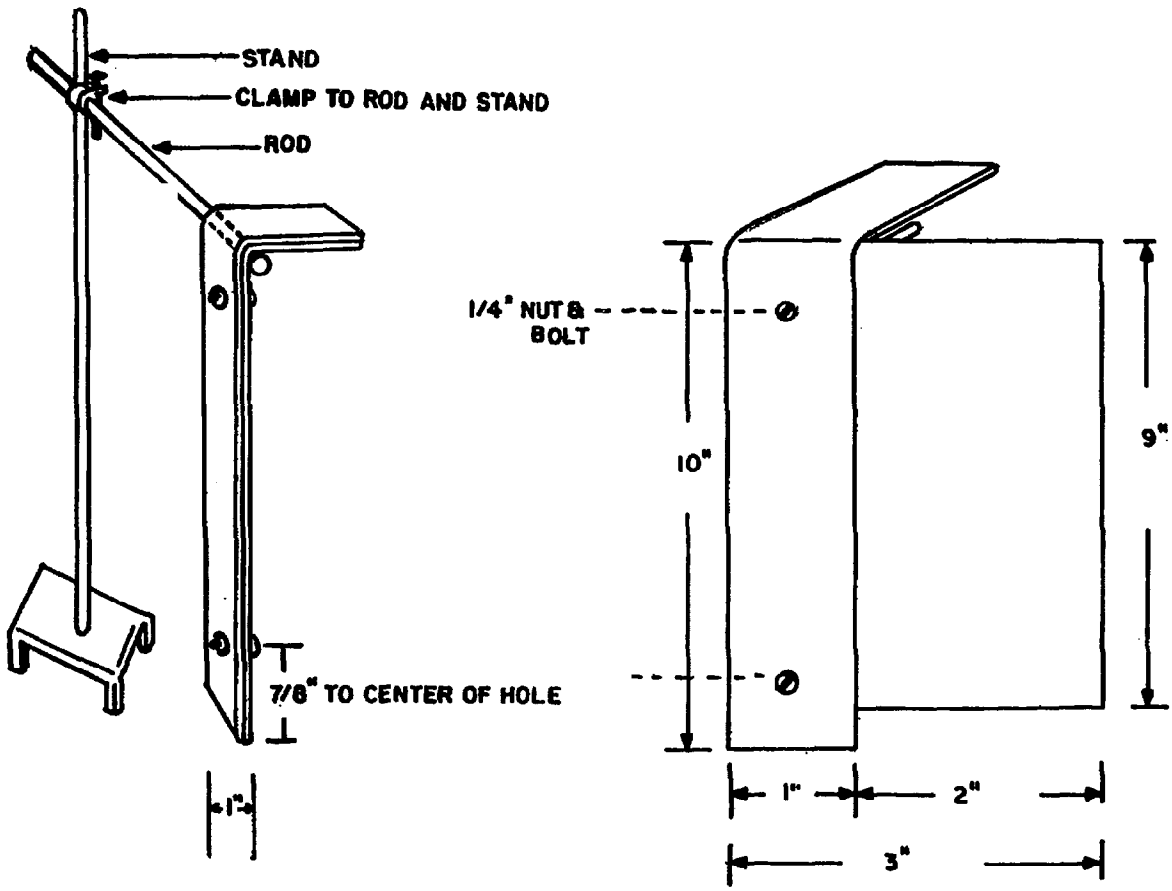


FIGURE 1. APPARATUS FOR DETERMINING BURNING RATE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:14Z
Check the source to verify that this is the current version before use.